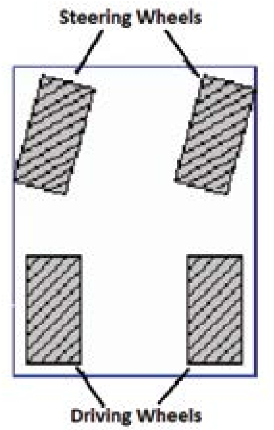
In differential type drive, zero radius of turning can be achieved by rotating the wheels at same rate in opposite direction. Another point to be noted here is that this design requires the use of two independent motors, and both of which are used in both rotational and translational motion.

3.2.2 Car type design



This design makes use of four wheels, two of them at the front for steering, and the rest two at the back to provide forward/ backward motion. This design type is very common in real world, and almost all the four wheeled vehicles make use of this design. However, this design type is fairly complex and thus is generally avoided in basic robotics.

3.2.3 Skid steer drive

This type is very similar to the differential drive design type. There are multiple numbers of driving wheels present in this, and there is no need of pivot wheel. The left and right groups of wheels are driven independently. This type of drive is commonly utilized in military tanks. In robotics, we make use of this type of design when there is a need of high friction and grip on the surface, for example, they are widely used in wall climbing robots. Turning